ORF409 - PS2

Chapter - 3

(Q.2) $x_0 = 3$ ~~$x_1 = (5x_0+7) \% 200$~~

$x_1 = (5x_0+7) \% 200 = 22$

$x_2 = (5x_1+7) \% 200 = 117$

$x_3 = (5x_2+7) \% 200 = 192$

$x_4 = (5x_3+7) \% 200 = 967 \% 200 = 167$

$x_5 = (5x_4+7) \% 200 = 842 \% 200 = 42$

$x_6 = (5x_5+7) \% 200 = 217 \% 200 = 17$

$x_7 = (5x_6+7) \% 200 = (5 \times 17+7) \% 200 = (85+7) \% 200 = 92 \% 200 = 92$

$x_8 = (5 \times 92+7) \% 200 = 467 \% 200 = 67$

$x_9 = (5 \times 67+7) \% 200 = 342 \% 200 = 142$

$x_{10} = (5 \times 142+7) \% 200 = 717 \% 200 = 117$

ORF409 - PS2

Chapter - 3

(Q.2) $x_0 = 3$ ~~$+ x_1 = (5x_0+7) \% 200$~~

$x_1 = (5x_0+7) \% 200 = 22$

$x_2 = (5x_1+7) \% 200 = 117$

$x_3 = (5x_2+7) \% 200 = 192$

$x_4 = (5x_3+7) \% 200 = 967 \% 200 = 167$

$x_5 = (5x_4+7) \% 200 = 842 \% 200 = 42$

$x_6 = (5x_5+7) \% 200 = 217 \% 200 = 17$

$x_7 = (5x_6+7) \% 200 = (5 \times 17+7) \% 200 = (85+7) \% 200 = 92 \% 200 = 92$

$x_8 = (5 \times 92+7) \% 200 = 467 \% 200 = 67$

$x_9 = (5 \times 67+7) \% 200 = 342 \% 200 = 142$

$x_{10} = (5 \times 142+7) \% 200 = 717 \% 200 = 117$

Exact Answers:-

(Q.6) $\displaystyle\int_0^\infty \frac{x\,dx}{(1+x^2)^2}$

$\rightarrow t = x^2$

$\Rightarrow dt = 2x\,dx$

$\Rightarrow x\,dx = \frac{dt}{2}$

$x=0 \rightarrow t=0$

$x\to\infty \rightarrow t\to\infty$

$$= \int_0^\infty \frac{dt}{2(1+t)^2}$$

$$= \left[\frac{1}{2(1+t)}\right]_\infty^0$$

$$= \frac{1}{2}\left(\frac{1}{1}-\frac{1}{\infty}\right)$$

$$= \boxed{\frac{1}{2}}$$

(Q.7) we know that:-

$$\int_{-\infty}^{\infty} N(0,1)\,dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\,dx = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2\pi}$$

So, $\displaystyle\int_{-\infty}^{\infty} N(0,\tfrac{1}{2})\,dx = 1$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{1}{\sqrt{\frac{1}{2}}\sqrt{2\pi}} e^{-\frac{x^2}{2\cdot\frac{1}{2}}}\,dx = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}} e^{-x^2}\,dx = 1$$

$$\Rightarrow \boxed{\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}}$$

(Q.9) $\displaystyle\int_0^\infty\int_0^x e^{-x}e^{-y}\,dy\,dx$

$$= \int_0^\infty e^{-x}\left(\int_0^x e^{-y}\,dy\right)dx$$

$$= \int_0^\infty e^{-x}\left([e^{-y}]_x^0\right)dx$$

$$= \int_0^\infty e^{-x}(1-e^{-x})\,dx$$

$$= \int_0^\infty (e^{-x}-e^{-2x})\,dx$$

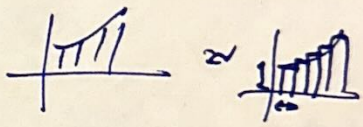
$$= [e^{-x}]_\infty^0 - \left[\frac{e^{-2x}}{2}\right]_\infty^0$$

$$= 1-0-\frac{1-0}{2}$$

$$= 1-\frac{1}{2}$$

$$= \boxed{\frac{1}{2}}$$

Simulation =



For (Q.6)

$\boxed{I = \int_0^\infty g(x)\,dx}$ $\left(g(x) = \frac{x}{(1+x^2)^2}\right)$

$\Rightarrow$ we want to bring $0-\infty$ to $0-1$

$\Rightarrow$ let $y = \frac{1}{x+1}$ $\Rightarrow dy = -\frac{1}{(x+1)^2}dx = -y^2\,dx$

$x\to 0 \Rightarrow y\to 1$, $x\to\infty \Rightarrow y\to 0$

$dx = -\frac{dy}{y^2}$

$$\Rightarrow I = -\int_1^0 \frac{g(\frac{1}{y}-1)}{y^2}\,dy = \int_0^1 \frac{g(\frac{1}{y}-1)}{y^2}\,dy$$

~~then, if $U \sim U[0,1]$, $I =$~~ $\boxed{= \int_0^1 h(y)\,dy}$

$$\boxed{h(y) = \frac{g(\frac{1}{y}-1)}{y^2} = \frac{\frac{1}{y}-1}{(1+(\frac{1}{y}-1)^2)^2 y^2}}$$

If $y \sim U[0,1]$, $I = E(h(Y)) \sim \sum_{i=1}^{K} \frac{1}{K}(h(Y_i))$

K large

(Q.7) Note that :-

$$e^{-(-x)^2} = e^{-(x)^2}$$

$\Rightarrow f(-x) = f(x)$

$\Rightarrow f$ symmetric about 0

So, $\int_{-\infty}^{\infty} e^{-x^2} dx$

$= 2\int_0^{\infty} e^{-x^2} dx$

Now we can convert $0-\infty$ to $0-1$

($x \to y$ requires y to be a monotonic f^n of x)

$\Rightarrow y = \frac{1}{x+1}$ | $x=0 \to y=1$, $x \to \infty \Rightarrow y \to 0$

So, $\boxed{I = 2\int_0^{\infty} g(x)\,dx}$ $(g(x) = e^{-x^2})$

$= 2\int_1^0 g\left(\frac{1}{y}-1\right) \cdot \frac{-(dy)}{y^2}$

$= 2\int_0^1 \frac{g\left(\frac{1}{y}-1\right)}{y^2} dy$

$= \boxed{2\int_0^1 h(y)\,dy}$

where $h(y) = \frac{g\left(\frac{1}{y}-1\right)}{y^2} = \boxed{\frac{e^{-\left(\frac{1}{y}-1\right)^2}}{y^2}}$

(Q.8) Note that :-

$$I = \int_0^{\infty}\int_0^{x} e^{-(x+y)} dy\,dx = \int_0^{\infty}\int_0^{\infty} I(y<x) e^{-(x+y)} dy\,dx$$

Now, again the same method :-

$u = \frac{1}{x+1}$, $v = \frac{1}{y+1} \Rightarrow \frac{-du}{u^2} = \frac{-dv}{v^2}$

$x: 0 \to \infty$, $u: 1 \to 0$; $y: 0 \to \infty$, $v: 1 \to 0$

So, $I = \int_1^0\int_1^0 I(u<v) e^{-\left(\frac{1}{u}-1+\frac{1}{v}-1\right)} \left(+\frac{du}{u^2}\right)\left(+\frac{dv}{v^2}\right)$

$= \int_0^1\int_0^1 I(u<v) \frac{e^{-\left(\frac{1}{u}+\frac{1}{v}-2\right)}}{u^2 v^2} dv\,du$

If we assume $u \perp\!\!\!\perp v$

$$\boxed{I = E_{U,V}\left[I(u<v) \frac{e^{-\left(\frac{1}{u}+\frac{1}{v}-2\right)}}{u^2 v^2}\right] \text{ where } U, V \sim U[0,1]}$$

And since $x \perp\!\!\!\perp y$ (no info given about dependence, hence indep)

$f(x) \perp\!\!\!\perp g(y)$ $\forall f, g$

$\Rightarrow \frac{1}{x+1} \perp\!\!\!\perp \frac{1}{y+1}$

$\Rightarrow u \perp\!\!\!\perp v$

$\Rightarrow$ Our assumption is valid.

(Q.10) ~~Cov(U, e^U)~~

Exact Answer :-

$Cov(U, e^U)$

$= E(Ue^U) - E(U)E(e^U)$

$= \int_0^1 x e^x \cdot 1 dx - \frac{1}{2}\int_0^1 e^x \cdot 1 dx$

$= (xe^x)_0^1 - \int_0^1 e^x dx - \frac{1}{2}(e-1)$

$= e - (e-1) - \frac{1}{2}(e-1)$

$= 1 - \frac{1}{2}(e-1)$

$= \frac{2 - e + 1}{2}$

$= \boxed{\frac{3-e}{2}}$

~~$= \frac{3-2.718}{2}$~~

~~$= \frac{0.282}{2}$~~

~~$= 0.141$~~

simulation

$Cov(U, e^U)$

$= \frac{\sum_{i=1}^{k} u_i e^{u_i}}{k} - \frac{\left(\sum_{i=1}^{k} u_i\right)}{k} \frac{\left(\sum_{i=1}^{k} e^{u_i}\right)}{k}$

(Q.12)

Let $U_1, U_2, \ldots \sim U[0,1]$

$N = \min\{n : \sum_{i=1}^{n} U_i > 1\}$

(d) $P(N = \alpha)$

$= P\left(\sum_{i=1}^{\alpha-1} U_i \le 1, \sum_{i=1}^{\alpha} U_i > 1\right)$

$= P\left(\sum_{i=1}^{\alpha-1} U_i \le 1\right) - P\left(\sum_{i=1}^{\alpha-1} U_i \le 1, \sum_{i=1}^{\alpha} U_i \le 1\right)$

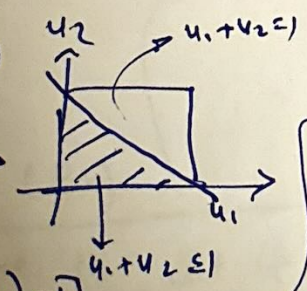
$= P\left(\sum_{i=1}^{\alpha-1} U_i \le 1\right) - P\left(\sum_{i=1}^{\alpha} U_i \le 1\right) P\left(\sum_{i=1}^{\alpha-1} U_i \le 1 \,\middle|\, \sum_{i=1}^{\alpha} U_i \le 1\right)$

$= P\left(\sum_{i=1}^{\alpha-1} U_i \le 1\right) - P\left(\sum_{i=1}^{\alpha} U_i \le 1\right)$

Now, $U \sim [0,1]$

If $Z = U_1 + U_2$, $f_Z(z) = P(Z = z)$

$= P(U_1 + U_2 = z)$



$F_Z(1) = P(Z \le 1) = P(U_1 + U_2 \le 1) = \frac{1}{2}$

for $Z = U_1 + U_2 + U_3$,

$F_Z(1) = P(U_1 + U_2 + U_3 \le 1)$

$= \int_0^1 \int_0^{1-u_1} \int_0^{1-u_1-u_2} du_3\, du_2\, du_1$

$= \int_0^1 \int_0^{1-u_1} (1 - u_1 - u_2)\, du_2\, du_1$

$= \int_0^1 \left[(1-u_1)u_2 - \frac{u_2^2}{2}\right]_0^{1-u_1} du_1$

$= \int_0^1 \frac{(1-u_1)^2}{2}\, du_1$

$= \left[\frac{(1-u_1)^3}{6}\right]_0^1 = \frac{1}{6}$

So, as seen the pattern is:-

$P\left(\sum_{i=1}^{k} U_i \le 1\right) = \frac{1}{k!}$

Hence, $P(N = \alpha) = \frac{1}{(\alpha-1)!} - \frac{1}{\alpha!}$

$= \frac{\alpha - 1}{\alpha!} \quad (\alpha \ge 2)$

$\Rightarrow E(N) = \sum_{\alpha=2}^{\infty} \frac{(\alpha-1)\cancel{\alpha}}{\cancel{\alpha!}(\alpha-1)!}$

$= \sum_{\alpha=2}^{\infty} \frac{1}{(\alpha-2)!}$

$= \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \ldots$

$= e^1\left(1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \ldots\right)$

$= \boxed{e}$

Chapter 4

(Q.3)

Order these probabilities in the decreasing order of their values so that the most likely one is kept at the top or at the first step of the (uniform) decision making process & so it is highest likely to exit in the first step itself which makes the process the most efficient.

$$H_i = \begin{cases} 0 & \text{w.p. } \frac{99}{100} \\ 1 & \text{w.p. } \frac{1}{100} \end{cases}$$

i.e. i^{th} card to be turned over $= \text{card}_i$

(Q.4) Let N = # hits (RV)

$$\Rightarrow \boxed{N = \sum_{i=1}^{100} H_i \quad \text{where } H_i = 1 \text{ if } i^{th} \text{ card gets a hit (success)}, \; = 0 \text{ otherwise}}$$

$$\Rightarrow E(N) = \sum_{i=1}^{100} E(H_i)$$

$$= \sum_{i=1}^{100} P(H_i = 1)$$

Now, i^{th} card can be placed anywhere in the sequence (of length 100) for $H_i = 1$ i.e. getting a hit, it should be in position i $\Rightarrow$ prob $= 1/100$

$$\Rightarrow \boxed{E(N) = \sum_{i=1}^{100} \frac{1}{100} = 1}$$

Now, to compute variances,

$$\text{var}(N) = E(N^2) - (E(N))^2$$

$$E(N^2) = \sum_{i=1}^{100} E[(H_i)^2] + \sum_{i \neq j} E(H_i H_j)$$

$$= \sum_{i=1}^{100} \frac{1}{100} + \sum_{i \neq j} E(H_i H_j)$$

$$= 1 + \sum_{i \neq j} E(H_i H_j) \quad -(*)$$

Now, $E(X_i X_j)$ a (Assuming $i \neq j$)

$= P$ (card$_i$ in i^{th} card/position & card$_j$ in j^{th} card/position)

$= P$(card$_i$ in i^{th} posn) P(card$_j$ in j^{th} posn / card$_i$ in i^{th} posn)

$= \frac{1}{100} \cdot \frac{1}{99}$ — ①

using ① in ⊛,

$$E(N^2) = 1 + \sum_{i \neq j} \frac{1}{100} \cdot \frac{1}{99}$$

$$= 1 + \frac{100 \cdot 99}{100 \cdot 99}$$

$$= 1 + 1$$

$$= 2$$

$\Rightarrow var(N) = E(N^2) - \cdot$

$= 2 - (1)^2$

$= 2 - 1$

$= \boxed{1}$

$\Rightarrow \boxed{var(N) = 1}$

So, exact answers i.e.

$\boxed{E(N) = 1,\ var(N) = 1}$

match with our estimates

(Q5)(b) By mathematical Induction,

$\boxed{\text{Base case}}$:- $n=1$ = [1]

⇒ The only possible permutation is [1] & that is there ✓

$\boxed{\text{Induction}}$: Assume that the claim holds for $n-1$.

i.e. The permutation obtained for $n-1$ (by the algo) is equally likely to be any of the $(n-1)!$ permutations.

$\boxed{\text{Proof}}$ (refer to ⊛ point at the end) — Let the permt realized for $n-1$ be :-

~~be $(P_1, P_2, \ldots, P_{n-1})$ (w) prob $\frac{1}{(n-1)!}$~~

step 1 (S1) $\underbrace{P_1^\circ, P_2^\circ, \ldots, P_{n-1}^\circ}$ ⇒ $L = P_1^\circ, P_2^\circ, \ldots, P_{n-1}^\circ$

$\boxed{\frac{1}{(n-1)!}}$

prob that we get some specific realization $P_1^\circ, P_2^\circ, \ldots, P_{n-1}^\circ$

(S2) (due to our assumption)

Next, $L' = L + [n] = P_1^\circ, P_2^\circ, \ldots, P_{n-1}^\circ, n$

(S3) Now, in the next step, we randomly pick an index i from $\{1, 2, \ldots, n\}$

$P(\text{index} = i) = \frac{1}{n}$

& then we swap $L[i]$ & $L[n]$ $(=n)$

⇒ The new sequence we get :-

$L'' = P_1^\circ, P_2^\circ, \ldots, P_{i-1}^\circ, n, P_{i+1}^\circ, \ldots, P_{n-1}^\circ, P_i^\circ$

~~$P(L'' \mid L(\text{or } P_1, P_2, \ldots, P_{n-1})) = \boxed{\frac{1}{n}}$~~

~~So, $P(L'', L) = P(L'' \mid L)\, P(L)$~~

~~$= \frac{1}{n} \frac{1}{(n-1)!}$~~

~~$= \frac{1}{n!}$~~ (further, whatever it generates is a valid sequence i.e. it only contains values from 1 to n ∈ Theoretically poss set of length-n seq)

~~So, $P(L'')$. Note that :-~~

~~if we show that $P(L'') = \frac{1}{n!}$, then~~

~~since L'' is generic / not a specific eg, L'' is an output of the algorithm formed using the algorithm, L'' is a placeholder for the outp~~

~~In general,~~

Note that :-

~~L'' is / represents~~ (from ~~step~~ (S1)) $\boxed{=}$ since the L (length $n-1$) sequence we assumed, is variabilized / generic & we just followed the algorithm there after,

L'' is / represents any possible output of the algorithm.

To that end, L'' is a generic variable storing (can store) all possible realizations (w) of the output $= [P_1^\circ, \ldots, P_{i-1}^\circ, n, P_{i+1}^\circ, \ldots, P_i^\circ]$

So, if we show that $P(L'') = \frac{1}{n!}$, that inherently means :-

$P(L'' = (w)) = \frac{1}{n!} \quad \forall w$

⇒ # w that this algorithm (thm) generates to make prob sum to 1 is $\frac{1}{p} = \frac{1}{\frac{1}{n!}} = n!$

⇒ [So, the algorithm generates $n!$ distinct sequences of length n, each equally likely].

[Since there are only $n!$ distinct such seq possible, the algo generates all possible sequences (of length n) with equal likelihood].

Now, $P(L'' = [P_1^\circ, P_2^\circ, \ldots, P_{i-1}^\circ, n, P_{i+1}^\circ, \ldots, P_{n-1}^\circ, P_i^\circ])$

as per the algo step

$\overset{S2 \to S3}{=} P(L' = [P_1^\circ, P_2^\circ, \ldots, P_{n-1}^\circ, n], \text{index} = i)$

$\overset{S1 \to S2}{=} P(L = [P_1^\circ, P_2^\circ, \ldots, P_{n-1}^\circ], \text{index} = i)$

$\overset{\because S1 \perp\!\!\!\perp S3}{=} P(L = [P_1^\circ, P_2^\circ, \ldots, P_{n-1}^\circ]) \cdot P(\text{index} = i)$

$= \frac{1}{(n-1)!} \cdot \frac{1}{n}$ $\to \because$ n possible indices

By induction assumption

$= \frac{1}{n!}$

— QED

(*) Note that in the (proof) part, L, L', L'' are the variables storing the values realized at S1, S2, S3. In general, they are not fixed & can take ~~possible / multiple~~ different values in different iterations.